Ridge and Lasso - Hyperparameters

CS109OA Introduction to Data Science

Pavlos Protopapas, Kevin Rader, and Chris Gumb

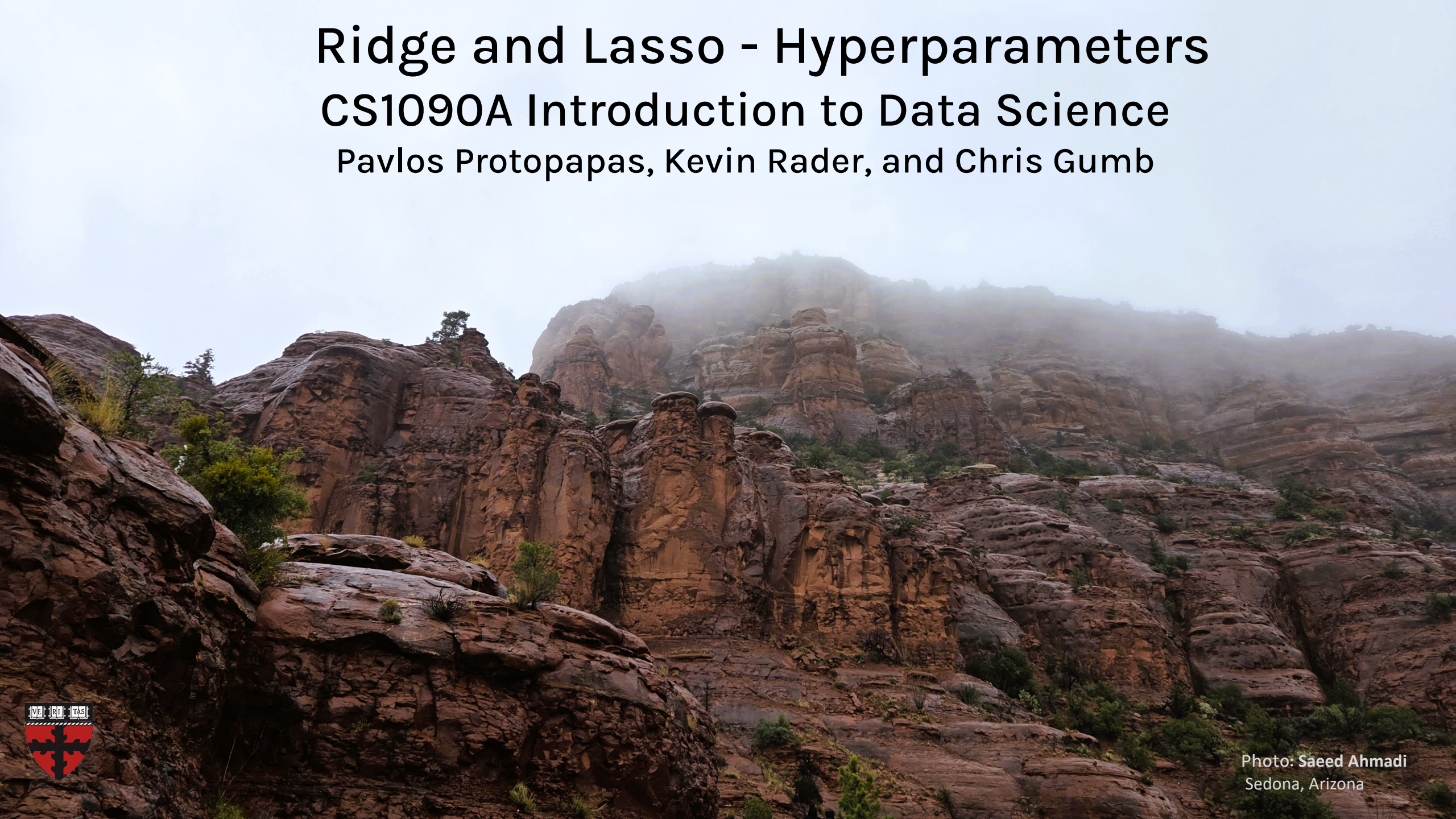


Photo: **Saeed Ahmadi**
Sedona, Arizona

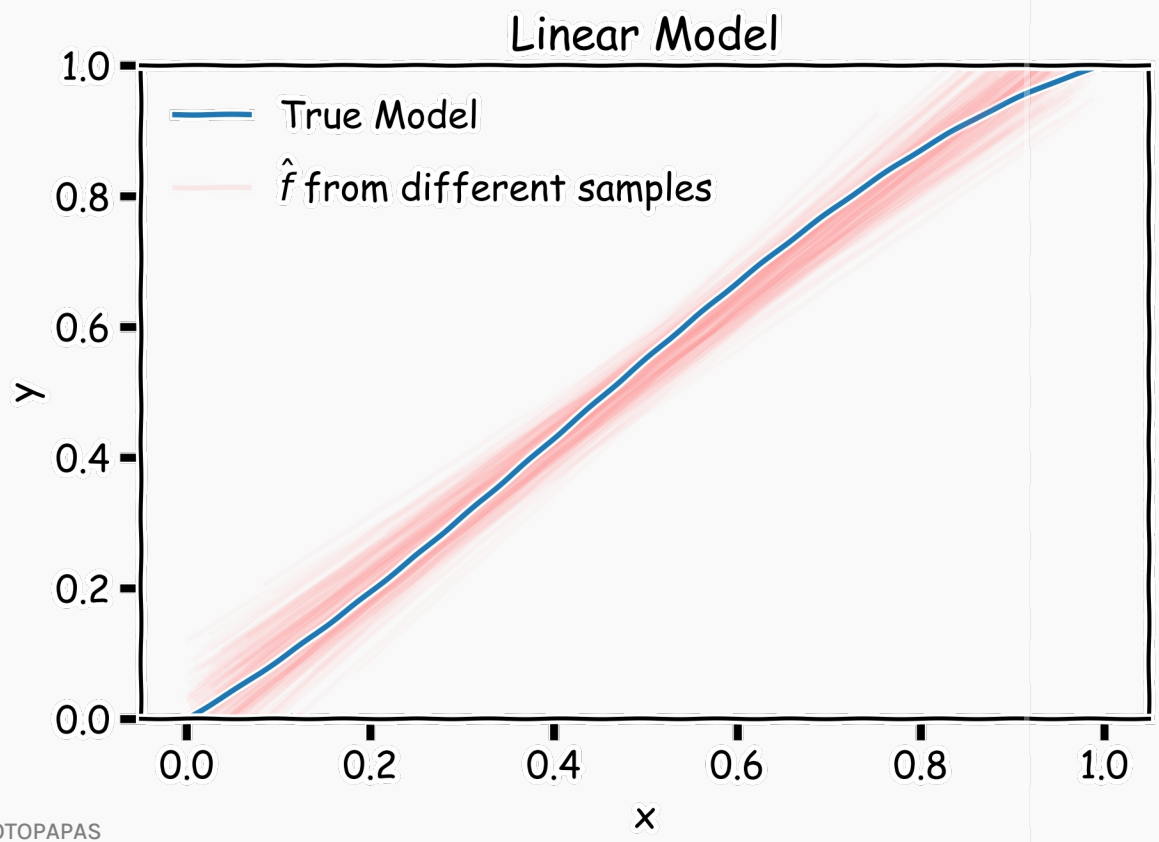
Outline

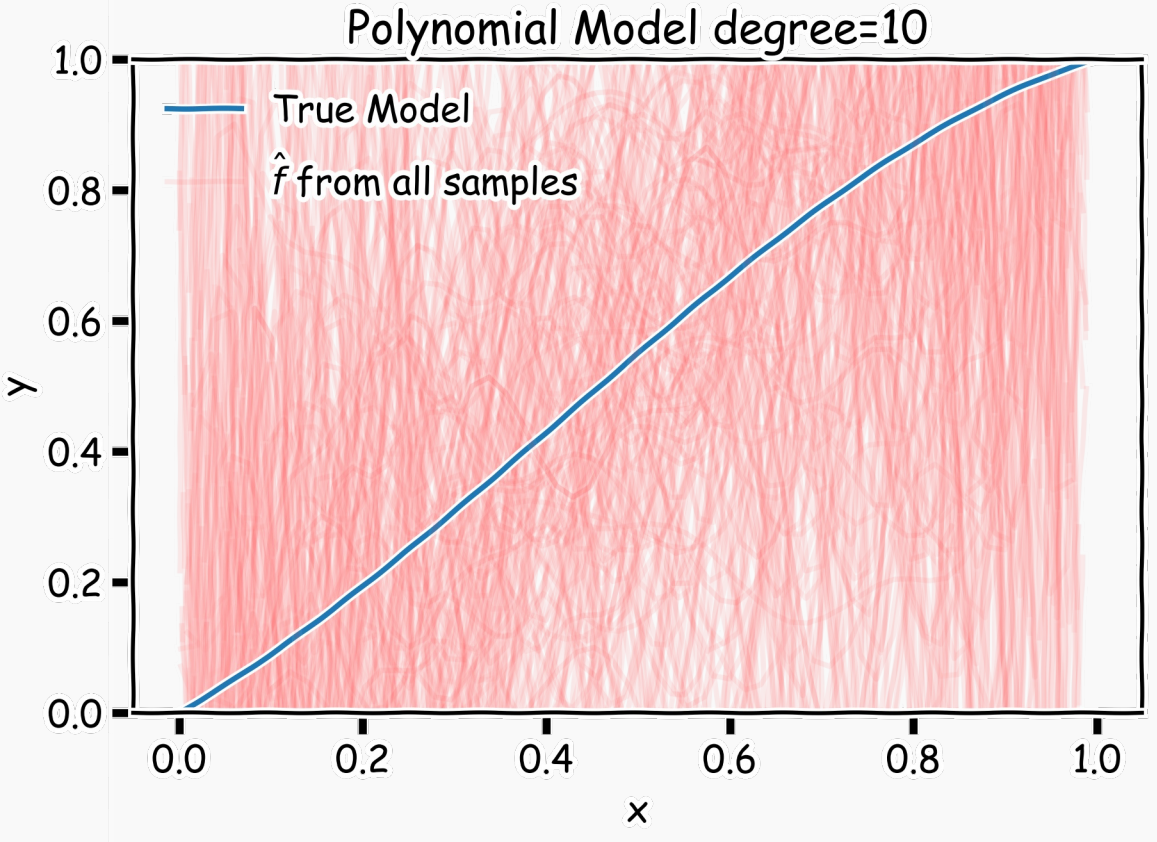
- Recap – Model Selection
- Generalization Error, Bias Variance Tradeoff
- **Regularization Techniques: Lasso, Ridge**

Bias vs Variance

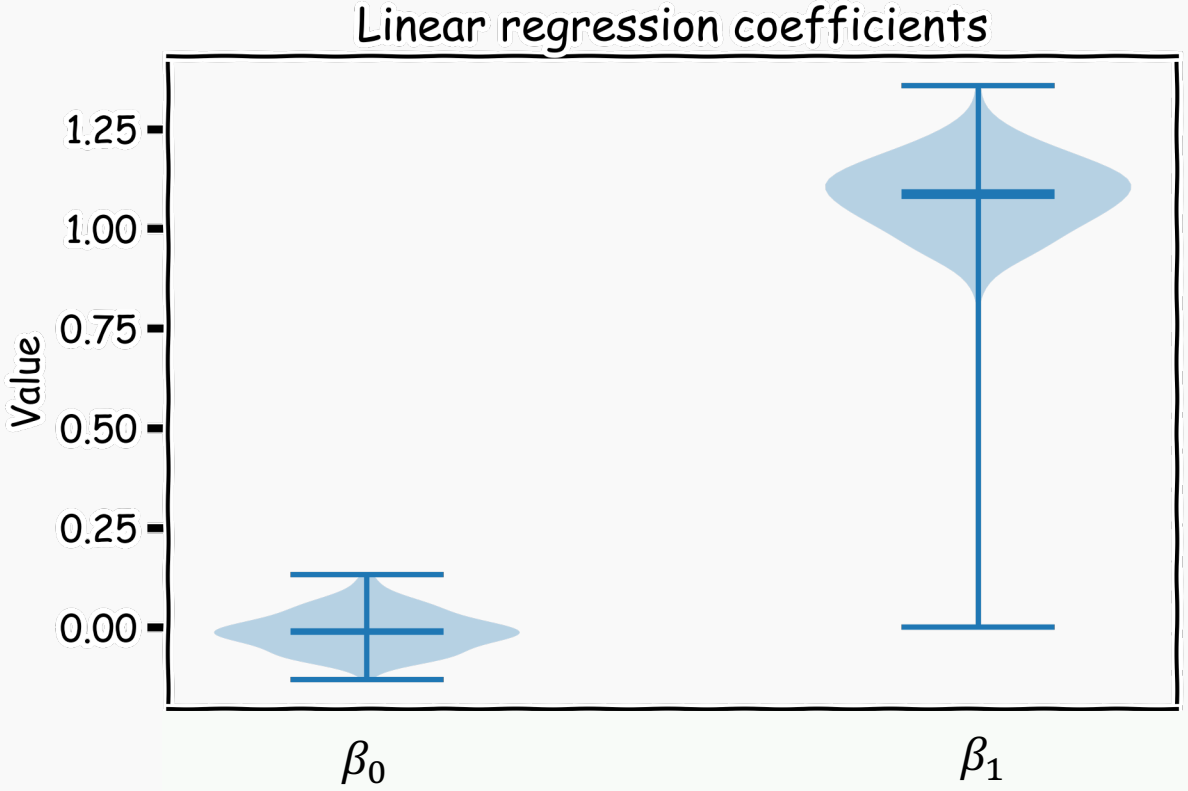
Left: 2000, best fit straight lines, each fitted on a different 20-point training set.

Right: Best-fit models using degree-10 polynomial





Bias vs Variance





Model Selection



Model selection is the application of a principled method to determine the complexity of the model, e.g., choosing a subset of predictors, choosing the degree of the polynomial model etc.

A strong m… …bid **overfitting**,

How do we discourage extreme values in the model parameters?

- there are…
  - the feature space has high dimensionality
  - the polynomial degree is too high
  - too many cross terms are considered
- the coefficients values are too **extreme**

Game Time



How would you discourage extreme values in the model parameters

Options:

A. Divide all model parameters by a large number

B. Make sure the causal relationship between predictors and response variable is true

C. Discard any model with model parameter value larger than 1

D. Penalize the model with a penalty that is proportional to the value its parameters

Game Time



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Regularization

What we want

Low model error

Minimize:

$$\frac{1}{n}\sum_{i=1}^{n}\left|y_i - \boldsymbol{\beta}^\top \boldsymbol{x}_i\right|^2$$

Discourage extreme values in model parameters

Minimize:

Regularization

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Discourage extreme values in model parameters

Minimize:

$$L_{reg} = \begin{cases} \sum_{j=1}^{J} \beta_j^2 \\ \sum_{j=1}^{J} |\beta_j| \end{cases}$$

Regularization



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Minimize:

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How do we combine these two objectives?

Regularization

What we want

Low mod… …e values in

Minimiz…

$$\frac{1}{n}\sum_{i=1}^{n}\left|y_i - \dots\right| \qquad \dots \beta_j^2 \qquad \dots |\beta_j|$$



Regularization

What we want

Low model error

Minimize:

Discourage extreme values in model parameters

Minimize:

$$\mathcal{L}_{REG} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \boldsymbol{\beta}^\top \boldsymbol{x}_i\right|^2 + L_{reg}$$

Regularization

What we want

Low model error

Discourage extreme values in model parameters

Minimiz mize:

λ is the **regularization parameter**. It controls the relative importance between model error and the regularization term

$$\mathcal{L}_{REG} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \boldsymbol{\beta}^\top \boldsymbol{x}_i\right|^2 + \lambda\ L_{reg}$$

Regularization

What we want

Low model error

Discourage extreme values in model parameters

mize:

$\lambda = 0$: equivalent to simple linear regression

$\lambda = \infty$: yields a model with $\beta's$ =0

$$\mathcal{L}_{REG} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \boldsymbol{\beta}^\top \boldsymbol{x}_i\right|^2 + \lambda\, L_{reg}$$

Regularization



What we want

Low model error

Minimize:

Discourage extreme values in model parameters

Minimize:



How do we determine λ?

$$\mathcal{L}_{REG} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \boldsymbol{\beta}^\top \boldsymbol{x}_i\right|^2 + \lambda L_{reg}$$

Regularization

What we want

Low model error

Minimize:

Discourage extreme values in model parameters

Minimize:

Cross Validation!

$$\mathcal{L}_{REG} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \boldsymbol{\beta}^\top \boldsymbol{x}_i\right|^2 + \lambda L_{reg}$$

Regularization: **LASSO** Regression

What we want

Low model error

Minimize:

Discourage extreme values in model parameters

Minimize:

Note that $\sum_{j=1}^{J} |\beta_j|$ is the l_1 norm of the vector $\boldsymbol{\beta}$

$$\mathcal{L}_{LASSO} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \boldsymbol{\beta}^\top \boldsymbol{x}_i\right|^2 + \lambda \sum_{j=1}^{J} |\beta_j|$$

Regularization: **LASSO** Regression



What we want

Low model error

Discourage extreme values in
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nize:

No need to regularize the bias, β_0
Why?

$$\mathcal{L}_{LASSO} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \boldsymbol{\beta}^\top \boldsymbol{x}_i\right|^2 + \lambda \sum_{j=1}^{J} |\beta_j|$$

Regularization: **LASSO** Regression

Lasso regression: minimize $\mathcal{L}_{LASSO}$ with respect to $\beta's$

$$\mathcal{L}_{LASSO} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \boldsymbol{\beta}^\top \boldsymbol{x}_i\right|^2 + \lambda\sum_{j=1}^{J}|\beta_j|$$

Regularization: Ridge Regression

Ridge regression: minimize $\mathcal{L}_{RIDGE}$ with re

Note that $\sum_{j=1}^{J} \beta_j^2$ is the L_2 norm square of the vector $\boldsymbol{\beta}$

$$\mathcal{L}_{RIDGE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \boldsymbol{\beta}^\top \boldsymbol{x}_i\right|^2 + \lambda \sum_{j=1}^{J} \beta_j^2$$

Regularization: Ridge Regression

Ridge regression: minimize $\mathcal{L}_{RIDGE}$ with respect to $\beta's$

$$\mathcal{L}_{RIDGE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \boldsymbol{\beta}^\top \boldsymbol{x}_i\right|^2 + \lambda \sum_{j=1}^{J} \beta_j^2$$

No need to regularize the bias, β_0, since it is not connected to the predictors.

Ridge regularization with only **validation** : step by step

For ridge regression there exist an analytical solution for the coefficients:

$$\hat{\beta}_{Ridge}(\lambda) = \left(\mathrm{X}^{\mathrm{T}}\mathrm{X} + \lambda I\right)^{-1} X^T Y$$



So, what are the steps to determine λ using validation?

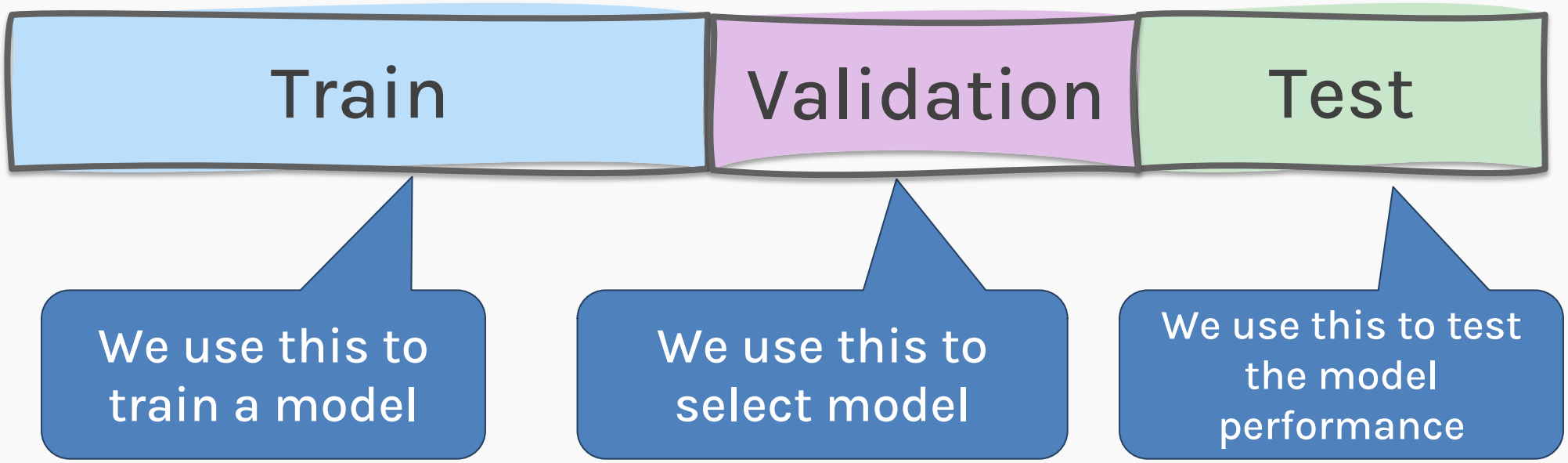
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Step 1 : Split Data

Split data into train, validation and test data.



Ridge regularization with only **validation** : step by step

Step 2 : Select a range of possible λ values

λ_{min}	λ_{min-1}	....	λ_{max-1}	λ_{max}

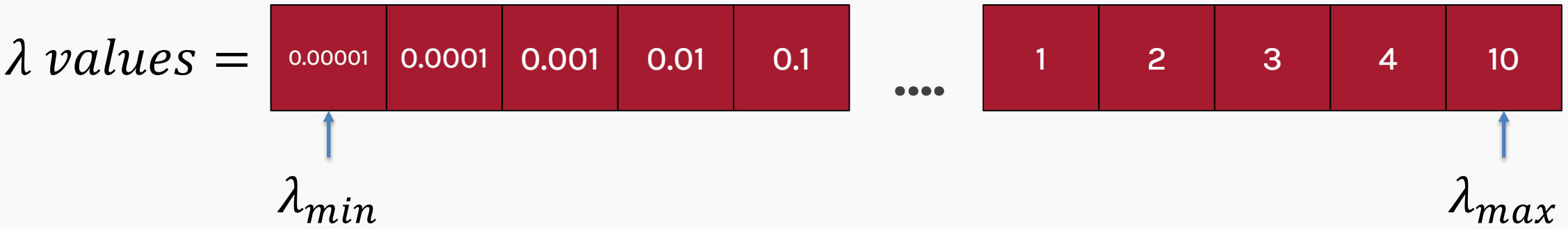


The values of λ are not fixed—you can choose the range yourself based on the problem.

Ridge regularization with only **validation** : step by step

Step 2 : Select a range of possible λ values

As an example, we will take a range of values from 0.00001 to 10



Ridge regularization with only **validation** : step by step

Step 3 : Determine β

Train

For each λ value, we determine $\beta_{Ridge}(\lambda)$ using train data

$\lambda\ values =$ 0.00001 | 0.0001 | 0.001 | 0.01 | 0.1 .... 1 | 2 | 3 | 4 | 10

$$\beta_{Ridge}(\lambda) = \left(\mathrm{X}^{\mathrm{T}}\mathrm{X} + \lambda I\right)^{-1} X^T Y$$

$\beta\ values =$

−1.401	−1.341	−1.665	4.702	−3.236	0.661	4.865	2.652	5.652	9.516

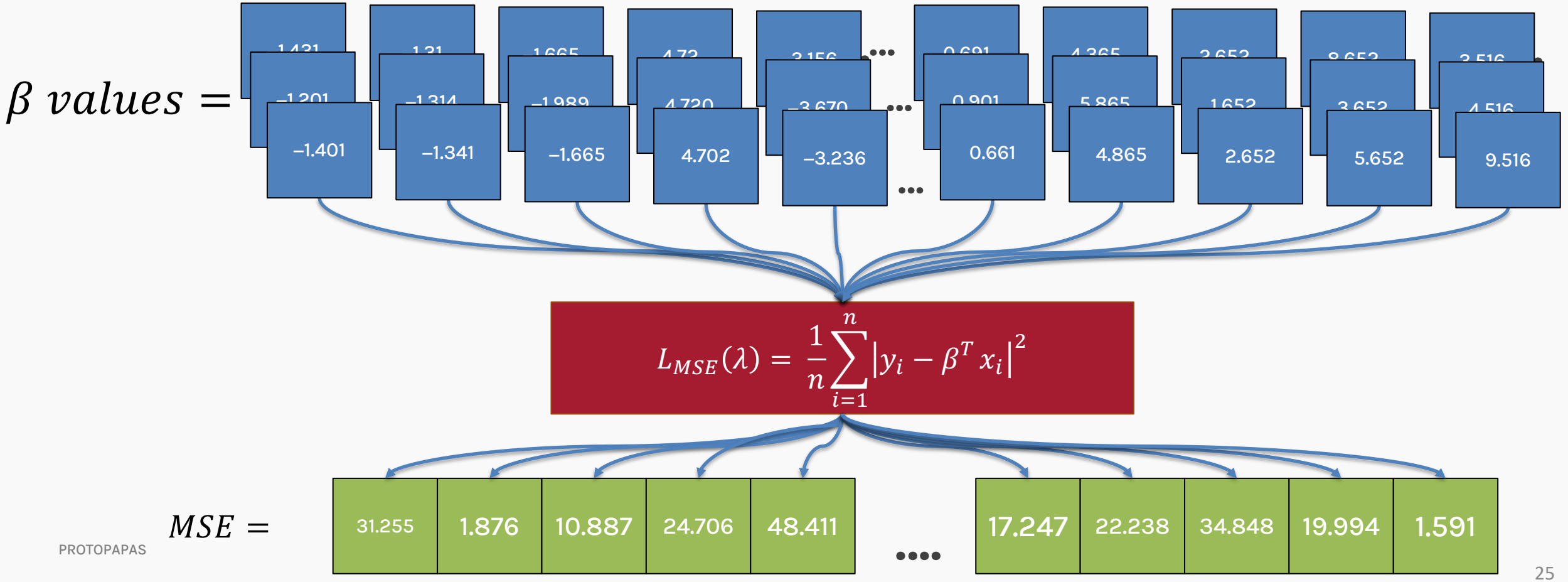
Keep in mind that each λ value gives us a different vector of β coefficients.

Ridge regularization with only **validation** : step by step

Validation

Step 4 : Record the $L_{MSE}(\lambda)$

For each vector of β coefficients, we record $L_{MSE}(\lambda)$ using the validation data



Ridge regularization with only **validation** : step by step

Step 5 : Select the λ_{Ridge}

Validation

Select the λ that minimizes the MSE loss on the validation data.

$\lambda\ values =$

0.00001	0.0001	0.001	0.01	0.1	....	1	2	3	4	10

$MSE\ =$

31.255	1.876	10.887	24.706	48.411	....	17.247	22.238	34.848	19.994	1.591

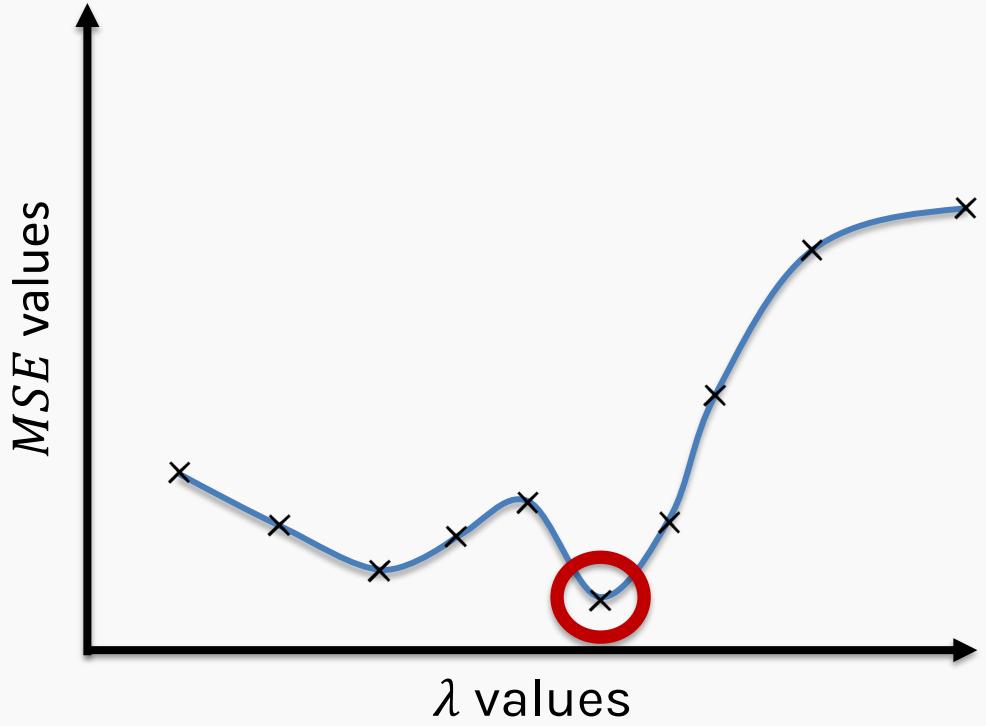
Let's visualize and see it!

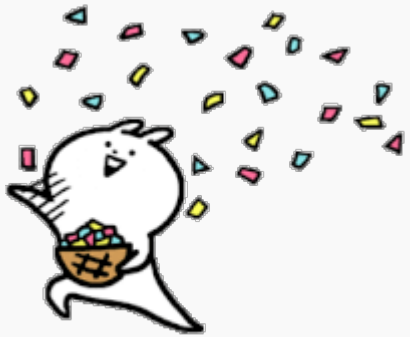
Ridge regularization with only **validation** : step by step

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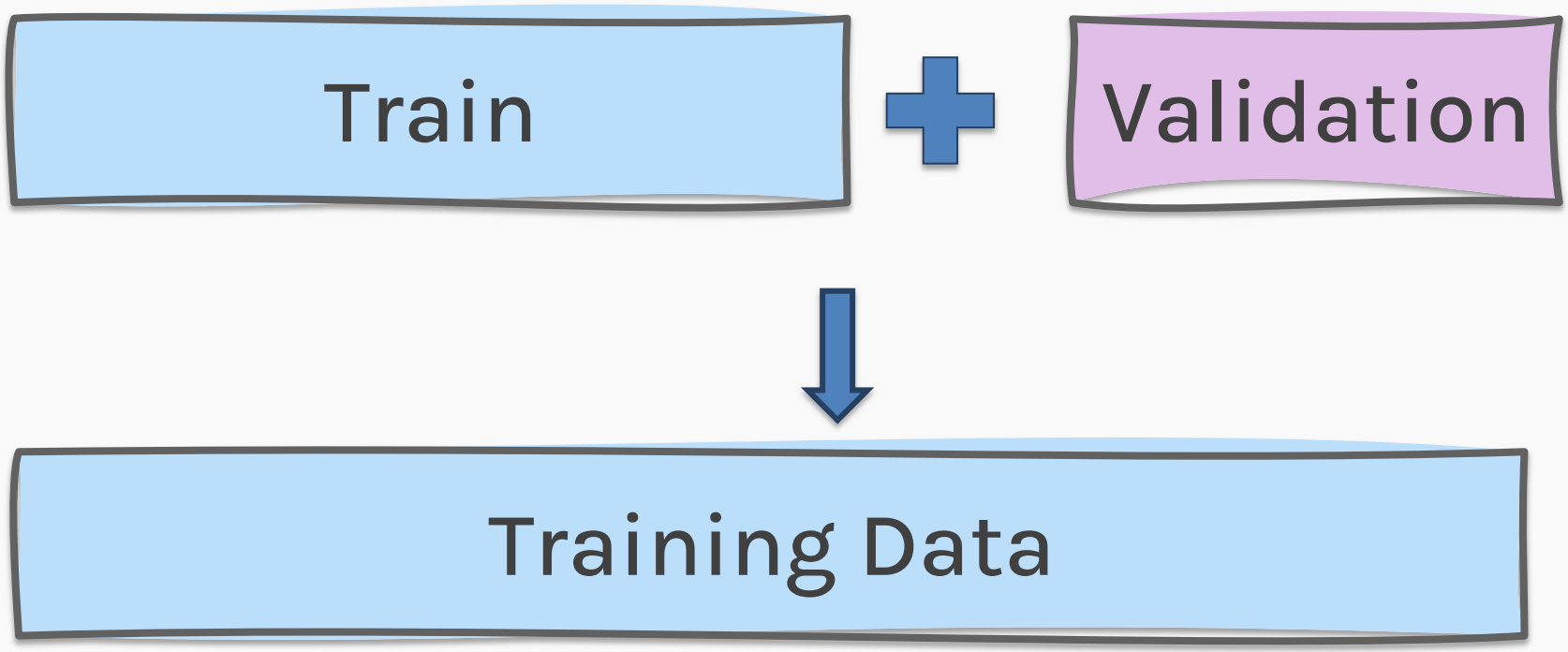




Ridge regularization with only **validation** : step by step

Step 6 : Refit the model

Refit the model using both train and validation data using λ_{Ridge}.

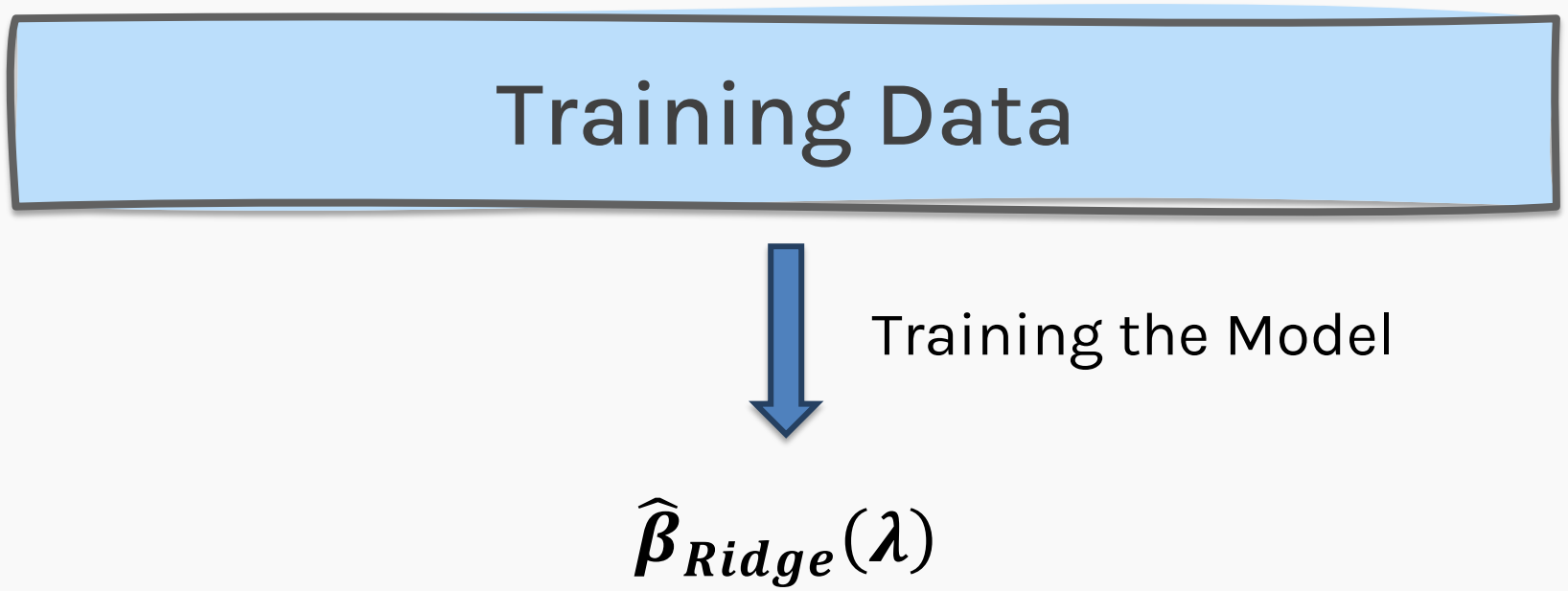


Ridge regularization with only **validation** : step by step

Step 6 : Refit the model

Refit the model using both train and validation data using $\boldsymbol{\lambda_{Ridge}}$.

This gives us $\widehat{\boldsymbol{\beta}}_{\boldsymbol{Ridge}}(\boldsymbol{\lambda})$



Ridge regularization with only **validation** : step by step

Step 7 : Record MSE/R2

Test

Report MSE or R2 on test data given the $\widehat{\boldsymbol{\beta}}_{\boldsymbol{Ridge}}(\boldsymbol{\lambda})$



Lasso regularization with only **validation** : step by step

For Lasso regression, there is **no** analytical solution for the coefficients, so we use a **solver**.



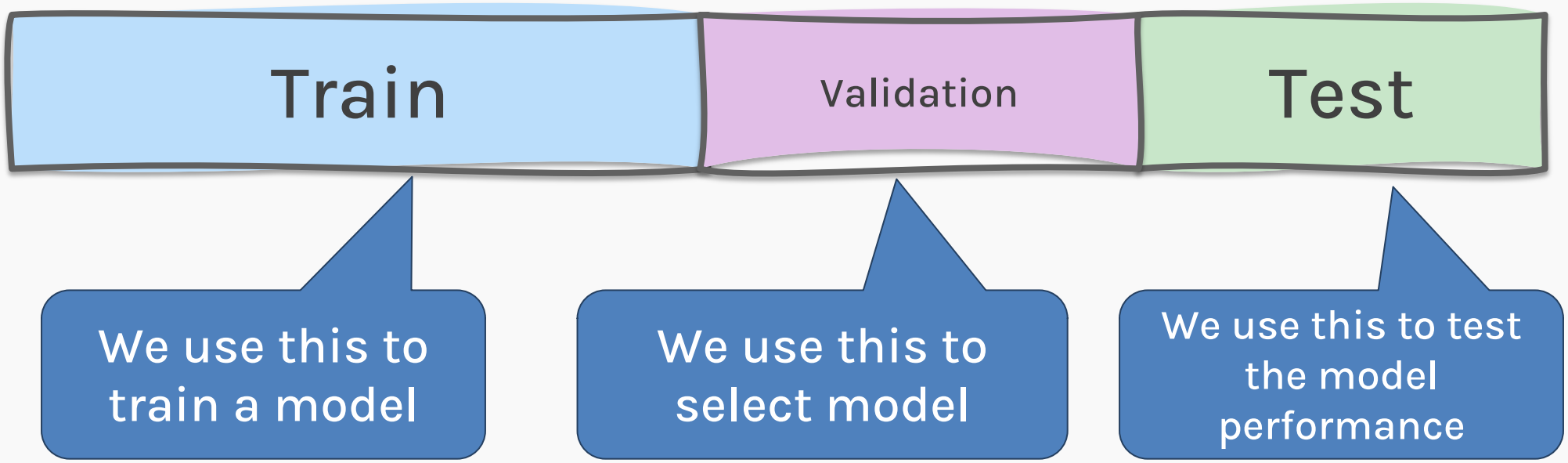
Now, the steps to determine λ using validation will still be the same!

Lasso regularization with only **validation** : step by step

For Lasso regression, there is **no** analytical solution for the coefficients, so we use a **solver**.

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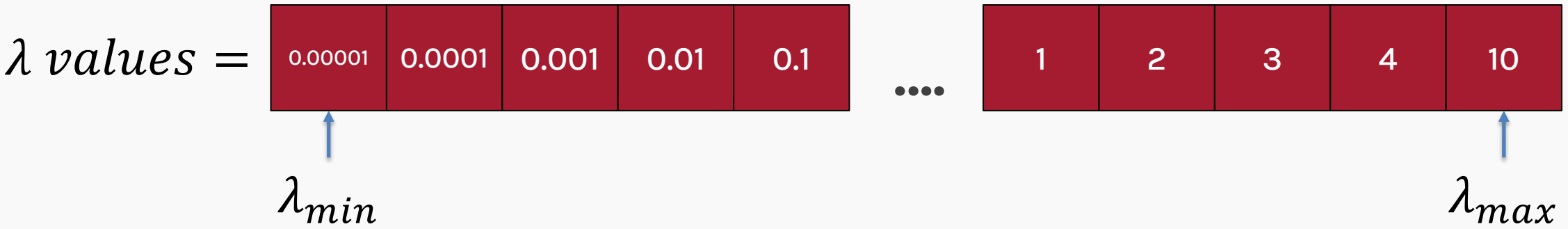


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Lasso regularization with only **validation** : step by step

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As an example, we will take a range of values from 0.00001 to 10

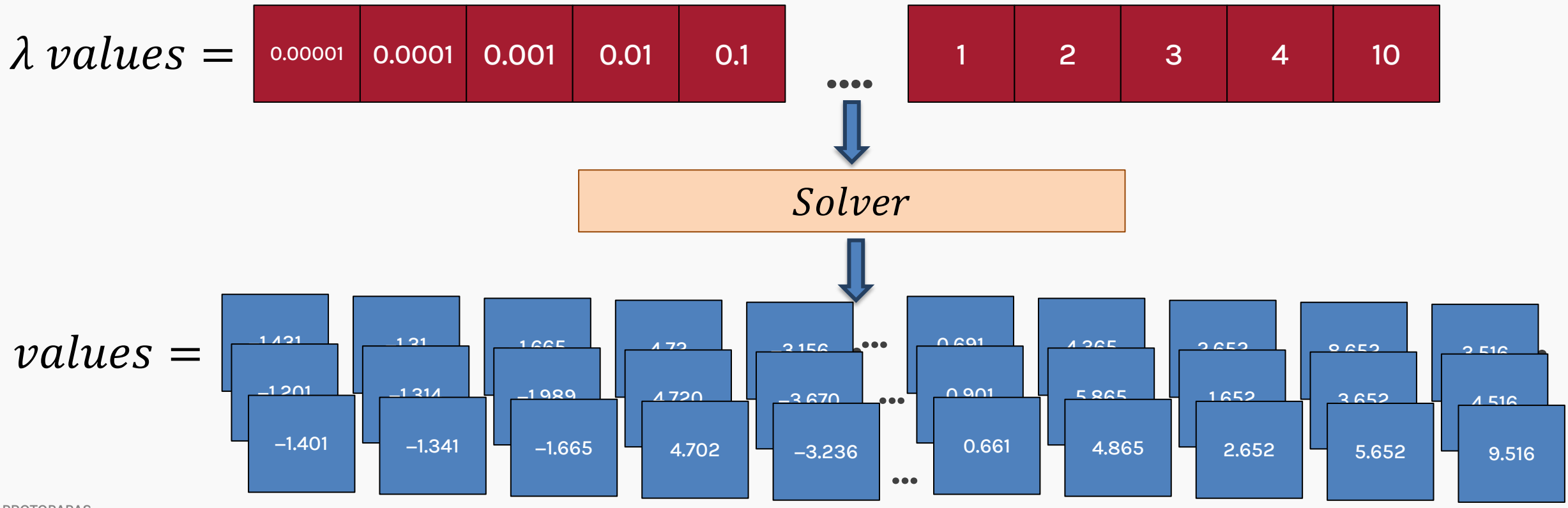


Lasso regularization with only **validation** : step by step

Step 3 : Determine β

Train

For each λ value, we determine $\beta_{Lasso}(\lambda)$ using train data



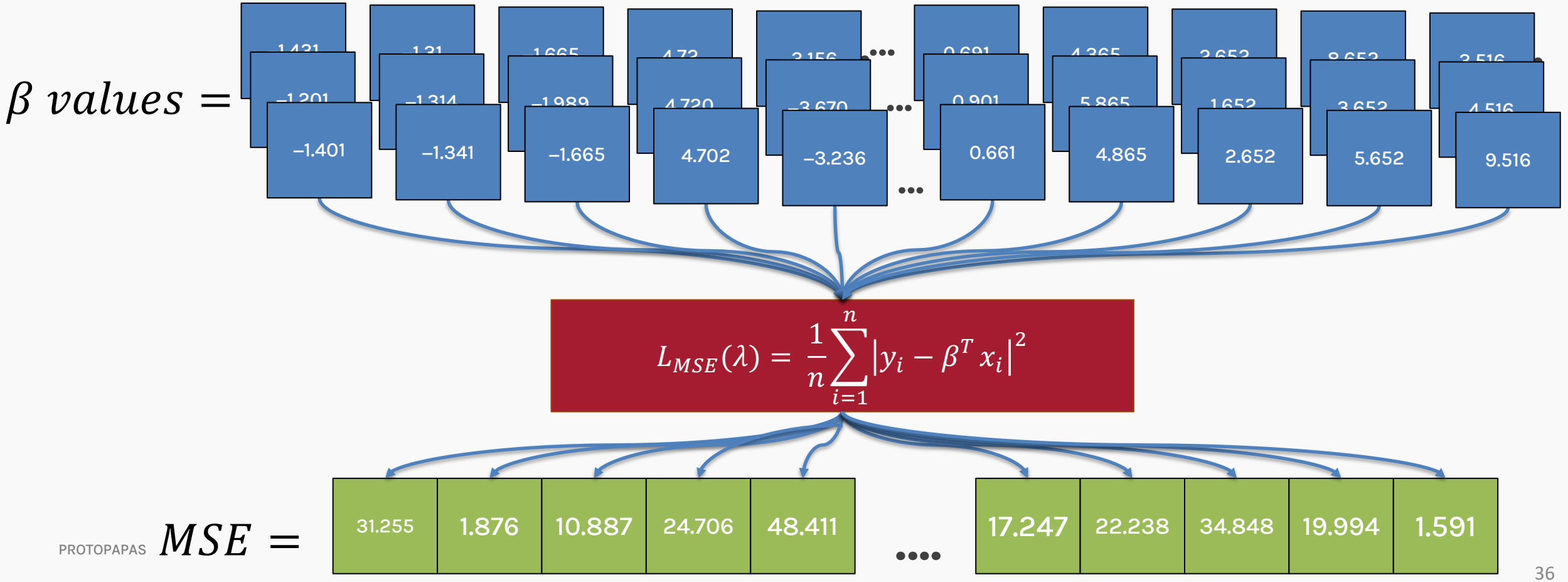
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Lasso regularization with only **validation** : step by step

Validation

Step 4 : Record the $L_{MSE}(\lambda)$

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Lasso regularization with only **validation** : step by step

Step 5 : Select the λ_{Lasso}

Validation

Select the λ that minimizes the MSE loss on the validation data.

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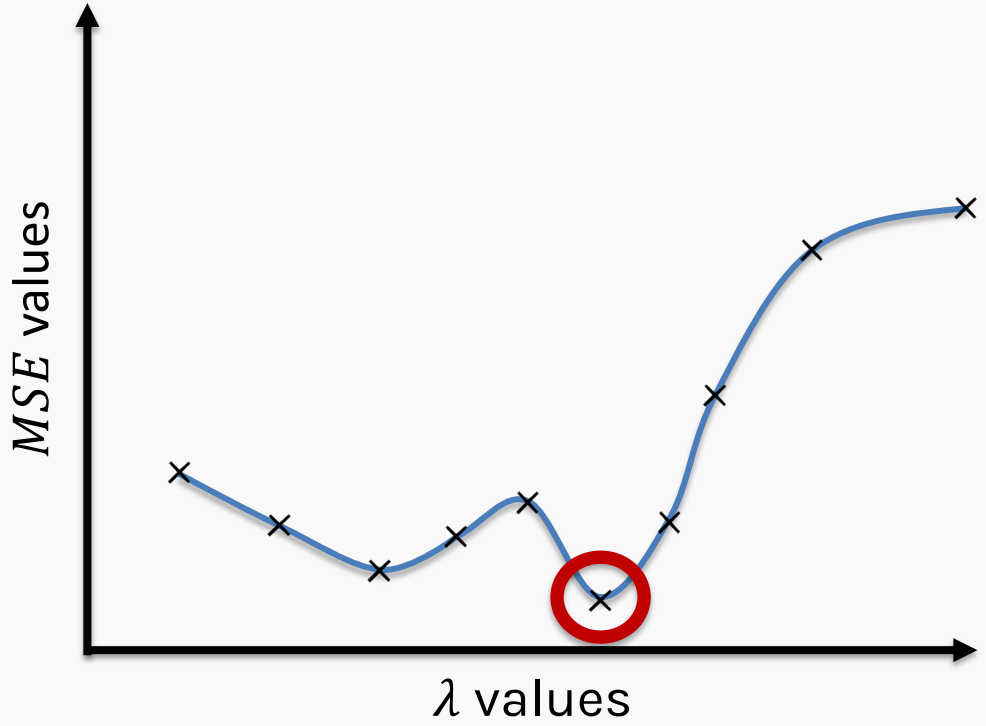
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Lasso regularization with only **validation** : step by step

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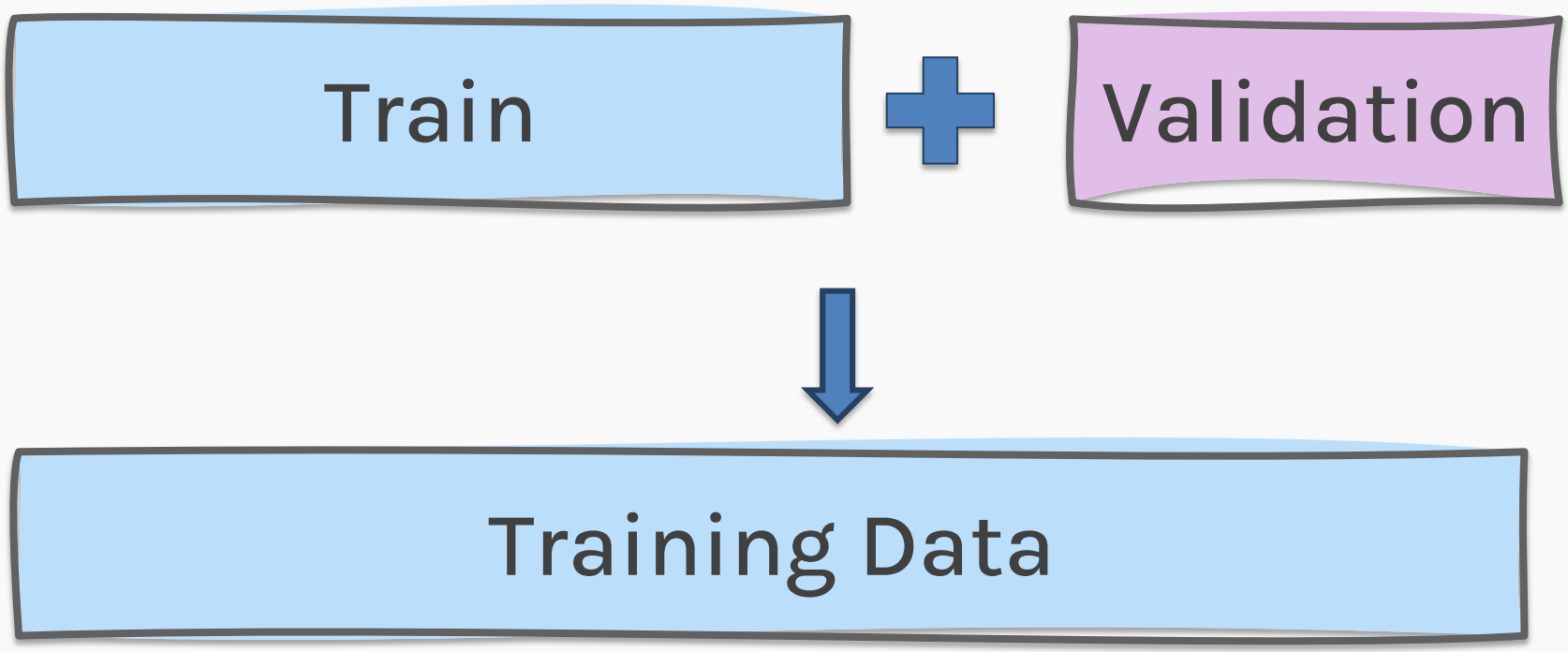




Lasso regularization with only **validation** : step by step

Step 6 : Refit the model

Refit the model using both train and validation data using λ_{Lasso}.

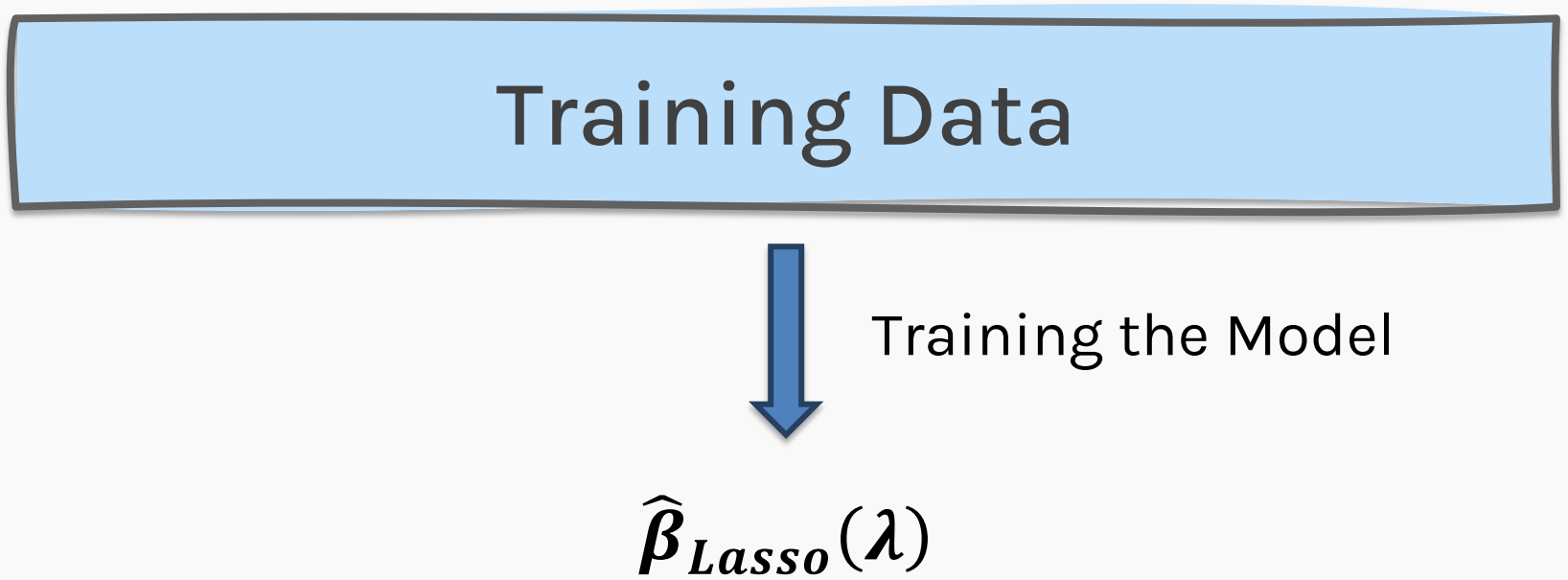


Lasso regularization with only **validation** : step by step

Step 6 : Refit the model

Refit the model using both train and validation data using λ_{Lasso}.

This gives us $\widehat{\boldsymbol{\beta}}_{Lasso}(\lambda)$



Lasso regularization with only **validation** : step by step

Step 7 : Record MSE/R2

Test

Report MSE or R2 on test data given the $\widehat{\boldsymbol{\beta}}_{\boldsymbol{Lasso}}(\boldsymbol{\lambda})$

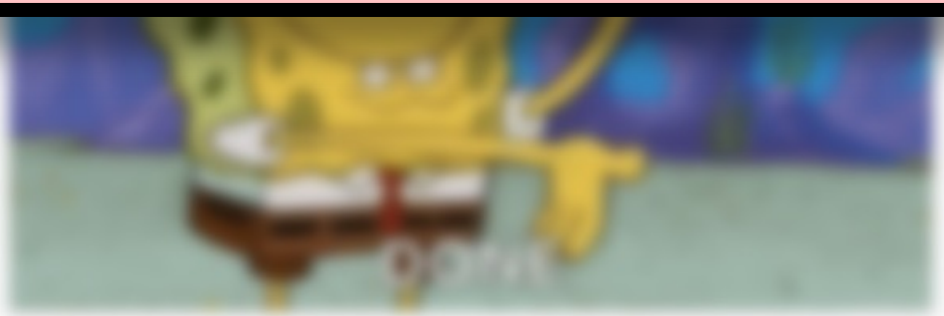


Lasso regularization with only validation : step by step

Step 7 : Record MSE/R2

Report MSE or R2 on test data given the $\hat{\beta}_{Lasso}(\lambda)$

Instead of relying on a single validation set, we can use cross-validation to get a more reliable estimate of the best λ



Ridge regularization with **CV**: step by step



Regularization with **CV**: step by step



Step 1 : Split Training Data

Split training data into K folds.



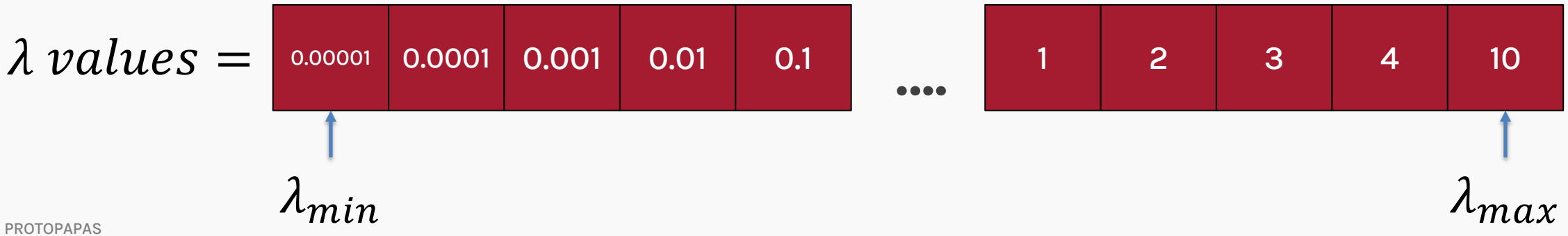
Regularization with **CV**: step by step

Step 2: Select a fold as Validation

Select one fold as validation and the rest as training data



Step 3: Select a range of possible λ values



Regularization with **CV**: step by step

Step 3 : Determine β

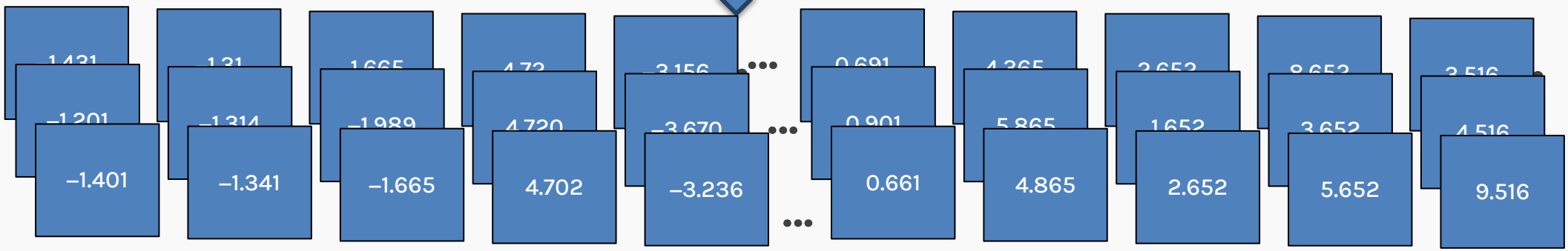


For each λ value, we determine $\beta_{Lasso/Ridge}(\lambda)$ using train data

$\lambda\ values =$ 0.00001 | 0.0001 | 0.001 | 0.01 | 0.1 .... 1 | 2 | 3 | 4 | 10

Ridge or Lasso Regularization

$\beta\ values =$



PROTOPAPAS

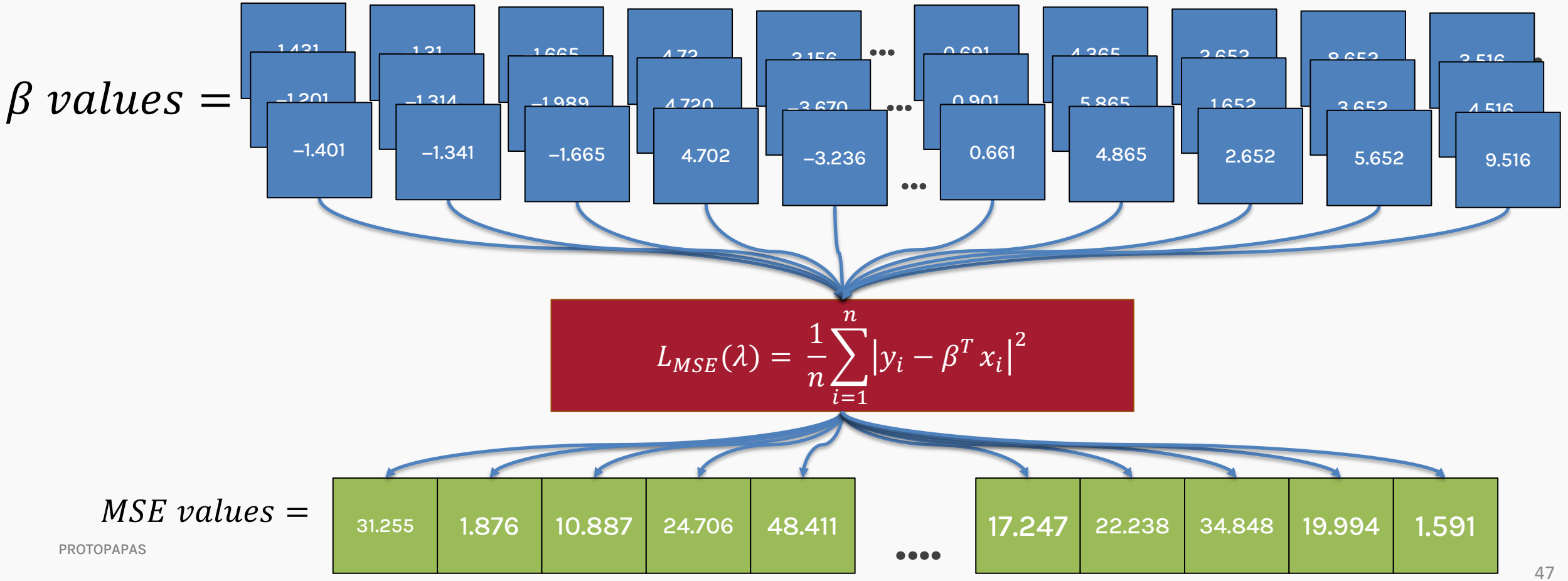
Keep in mind that each λ value gives us a different vector of β coefficients.

Regularization with **CV**: step by step

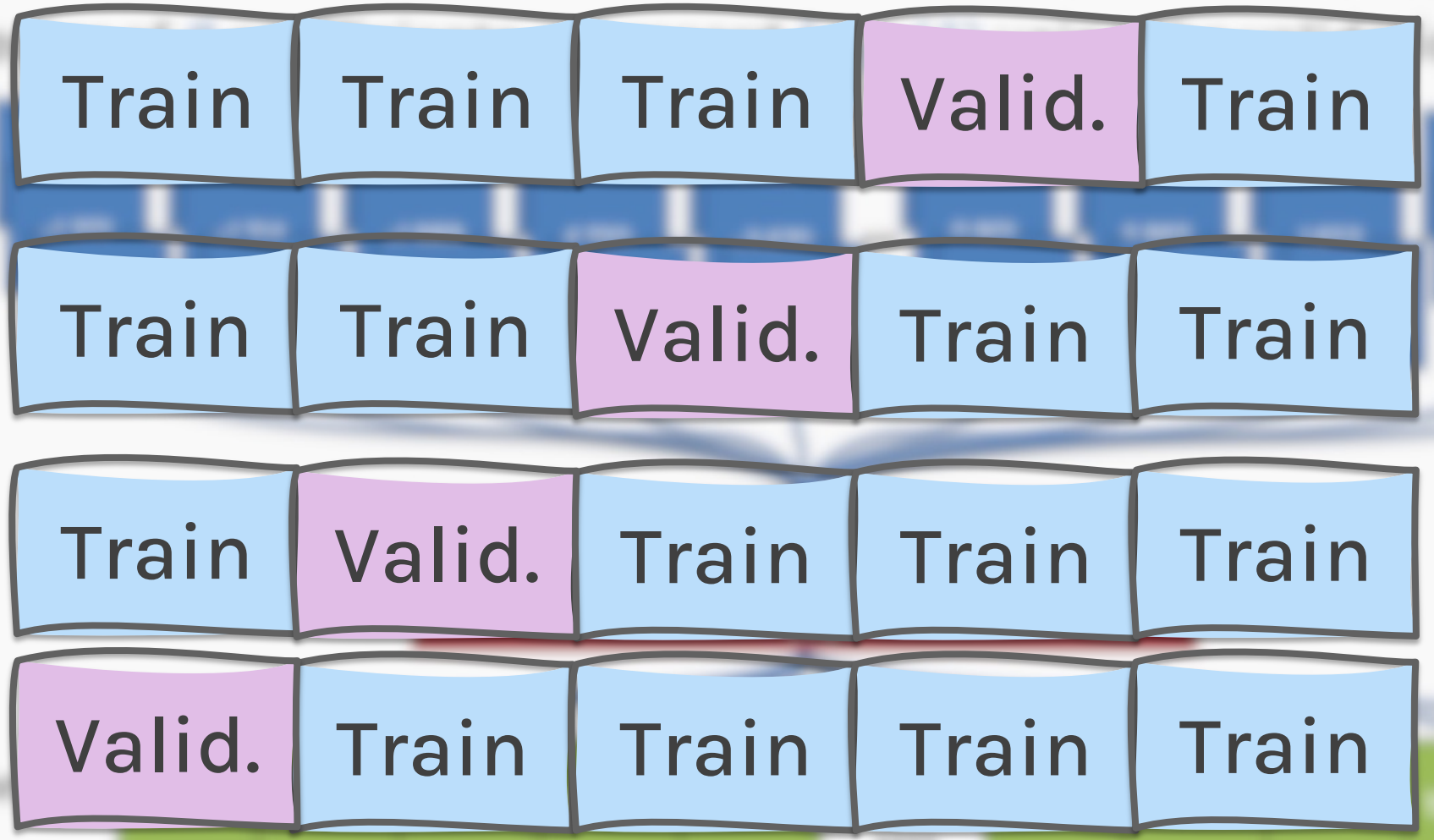
Validation

Step 4 : Record the $L_{MSE}(\lambda)$

For each vector of β coefficients, we record $L_{MSE}(\lambda)$ using the validation data



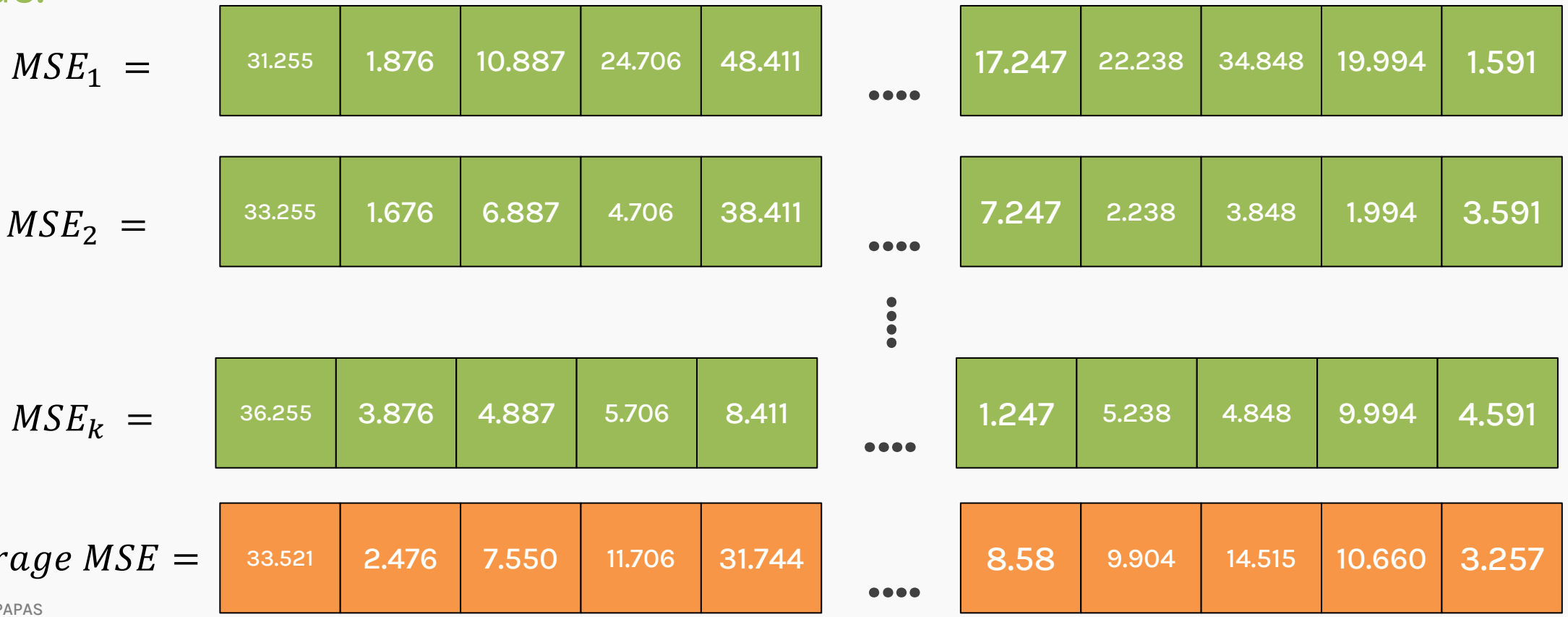
Repeat Step 2 to 4 for different values of k



Regularization with **CV**: step by step

Step 6: Average MSE, $\bar{L}_{MSE}(\lambda)$

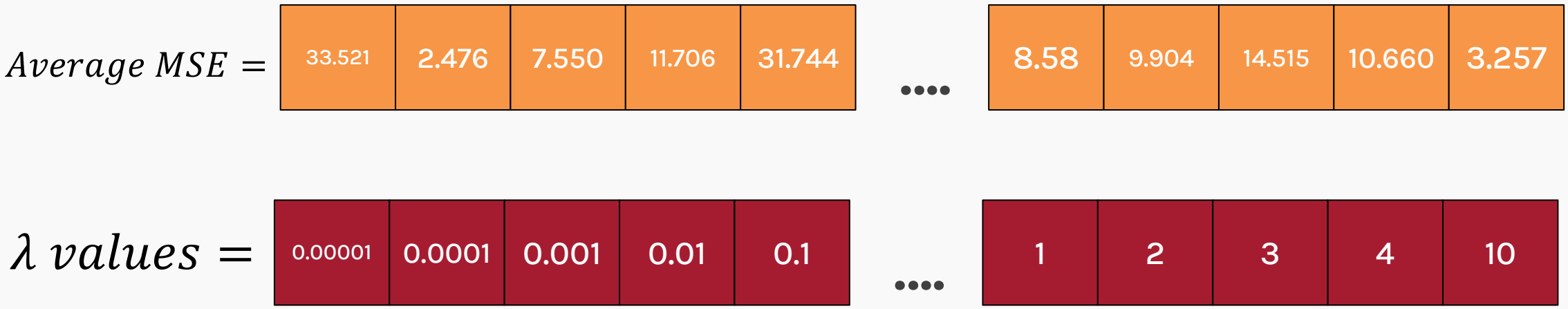
Calculate the average MSE, $\bar{L}_{MSE}(\lambda)$ for each λ by averaging $L_{MSE}(\lambda, k)$ over k. folds.



Regularization with **CV**: step by step

Step 7: Select λ

Find the λ that minimizes the $\bar{L}_{MSE}(\lambda)$

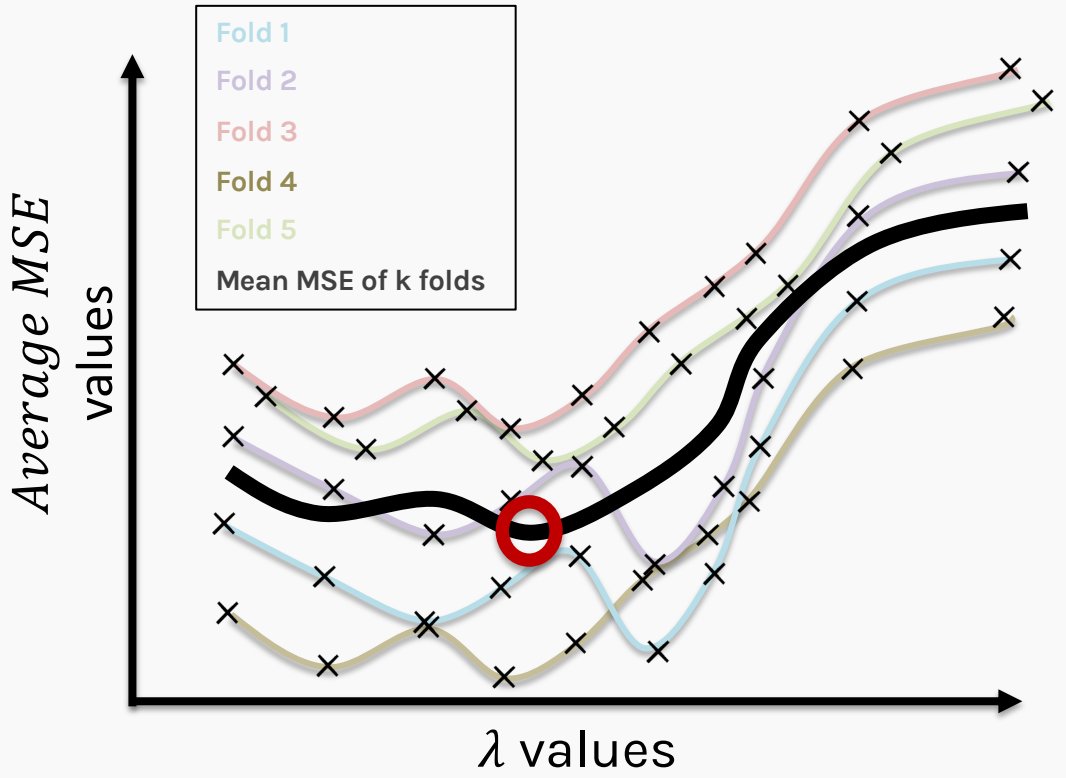


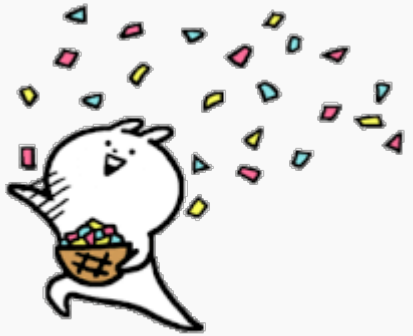
Let's visualize and see it!

Regularization with **CV**: step by step

Step 7: Select λ

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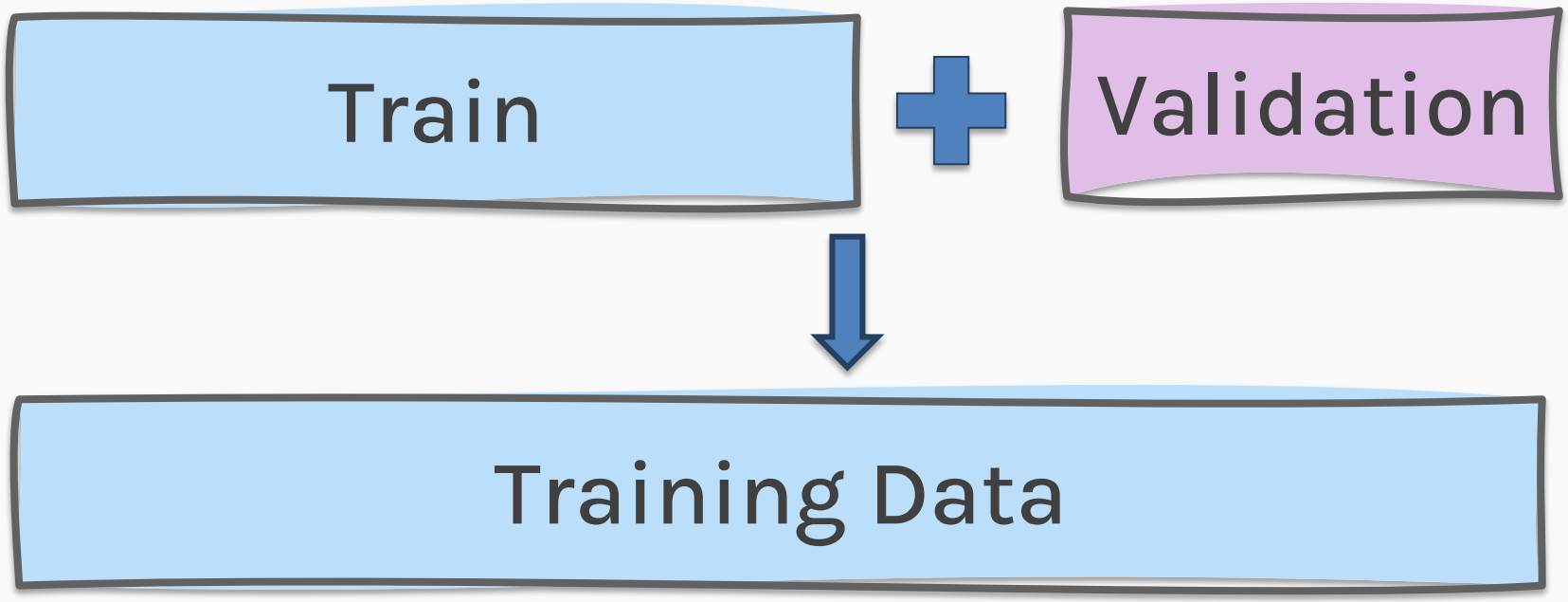




Regularization with **CV**: step by step

Step 8: Refit the model

Refit the model using both train and validation data using λ.

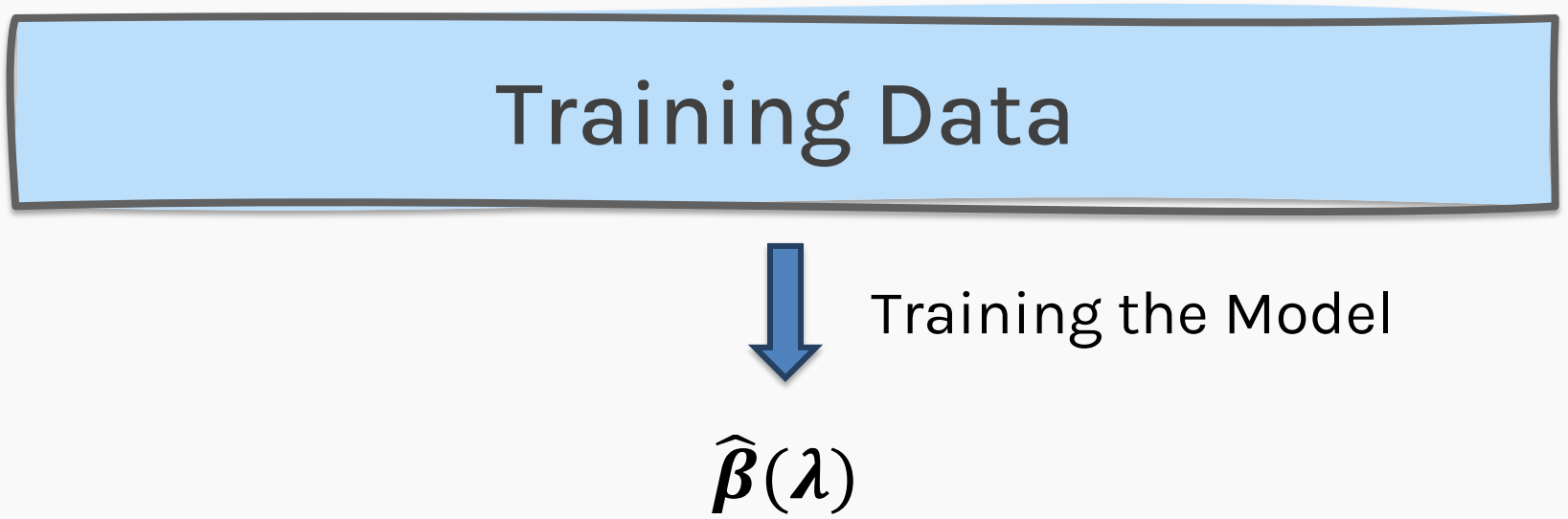


Regularization with **CV**: step by step

Step 8: Refit the model

Refit the model using both train and validation data using λ.

This gives us $\widehat{\boldsymbol{\beta}}(\lambda)$

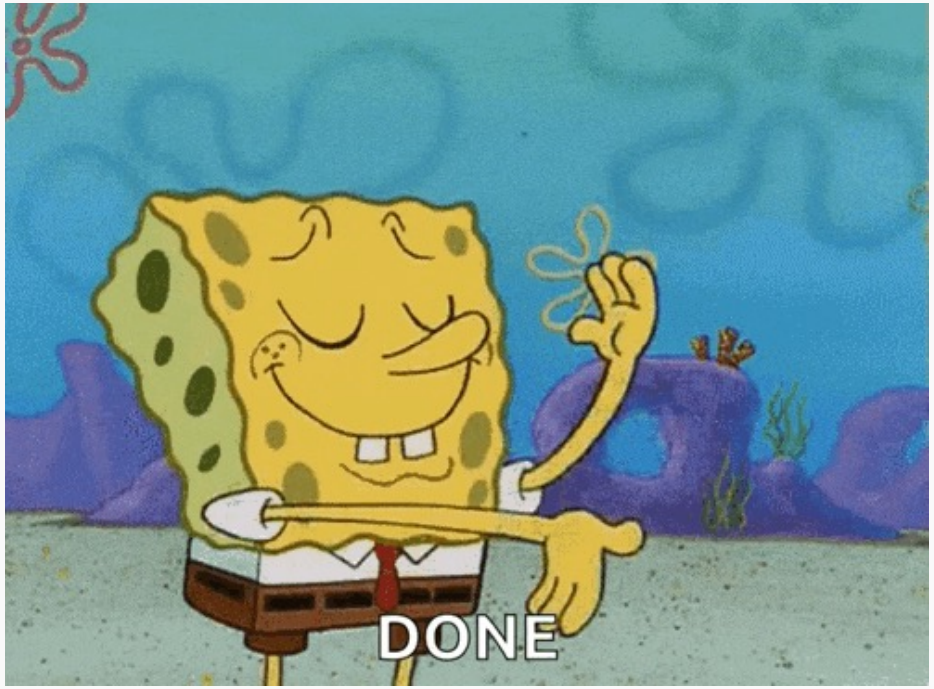


Regularization with **CV**: step by step

Step 9 : Record MSE/R2



Report MSE or R2 on test data given the $\widehat{\boldsymbol{\beta}}(\boldsymbol{\lambda})$



Ridge regularization with only **validation** : step by step

For ridge regression there exist an analytical solution for the coefficients:

$$\hat{\beta}_{Ridge}(\lambda) = \left(\mathrm{X}^{\mathrm{T}}\mathrm{X} + \lambda I\right)^{-1} X^T Y$$

1. split data into $\{\{X,Y\}_{train}, \{X,Y\}_{validation}, \{X,Y\}_{test}\}$
2. for λ in $\{\lambda_{min}, \dots \lambda_{max}\}$:
    1. determine the β that minimizes the L_{ridge}, $\beta_{Ridge}(\lambda) = \left(\mathrm{X}^{\mathrm{T}}\mathrm{X} + \lambda I\right)^{-1} X^T Y$, using the train data.
    2. record $L_{MSE}(\lambda)$ using validation data.
3. select the λ that minimizes the *MSE* loss on the validation data,

$$\lambda_{ridge} = \mathrm{argmin}_{\lambda}\, L_{MSE}(\lambda)$$

4. Refit the model using both train and validation data, $\{\{X,Y\}_{train}, \{X,Y\}_{validation}\}$, now using λ_{ridge}, resulting to $\hat{\beta}_{ridge}(\lambda_{ridge})$
5. Report MSE or R^2 on $\{X,Y\}_{test}$ given the $\hat{\beta}_{ridge}(\lambda_{ridge})$

Lasso regularization with **validation** only: step by step

For Lasso regression, there is **no** analytical solution for the coefficients, so we use a **solver**.

1. split data into $\{\{X,Y\}_{train}, \{X,Y\}_{validation}, \{X,Y\}_{test}\}$
2. for λ in $\{\lambda_{min}, \dots \lambda_{max}\}$:
   A. determine the β that minimizes the L_{lasso}, $\beta_{lasso}(\lambda)$, using the train data. **This is done using a solver.**
   B. record $L_{MSE}(\lambda)$ using the validation data.
3. select the λ that minimizes the ***MSE*** **loss** on the validation data,

$$\lambda_{lasso} = \text{argmin}_{\lambda}\, L_{MSE}(\lambda)$$

4. Refit the model using both train and validation data, $\{\{X,Y\}_{train}, \{X,Y\}_{validation}\}$, now using λ_{Lasso}, resulting to $\hat{\beta}_{lasso}(\lambda_{lasso})$
5. Report MSE or R^2 on $\{X,Y\}_{test}$ given the $\hat{\beta}_{lasso}(\lambda_{lasso})$

Ridge regularization with CV: step by step

	λ_1	λ_2	...	λ_n
k_1	L_{11}	L_{12}	..	...
k_2	L_{21}	...	..	...
...	..	...	..	...
k_n	...	...	...	...
E[]	$\bar{L}_1$	$\bar{L}_2$	...	$\bar{L}_n$

1. remove $\{X,Y\}_{test}$ from data
2. split the rest of data into K folds, $\{\{X,Y\}_{train}^{-k}, \{X,Y\}_{val}^{k}\}$
3. for k in $\{1,\dots,K\}$

   for λ in $\{\lambda_0,\dots,\lambda_n\}$:

   A. determine the β that minimizes the L_{ridge}, $\beta_{ridge}(\lambda,k) = \left(\mathrm{X^T X} + \lambda I\right)^{-1} X^T Y$, **using the train data of the fold, $\{X,Y\}_{train}^{-k}$.**

   B. record $L_{MSE}(\lambda,k)$ using the validation data of the fold $\{X,Y\}_{val}^{k}$

At this point we have a 2-D matrix, rows are for different k, and columns are for different λ values.

4. Calculate the average MSE, $\bar{L}_{MSE}(\lambda)$ the for each λ by averaging $L_{MSE}(\lambda,k)$ over k folds.
5. Find the λ that minimizes the $\bar{L}_{MSE}(\lambda)$, resulting to λ_{ridge}.
6. Refit the model using the full **training data, $\{\{X,Y\}_{train}, \{X,Y\}_{val}\}$, resulting to $\hat{\beta}_{ridge}(\lambda_{ridge})$**
7. report MSE or R^2 on $\{X,Y\}_{test}$ given the $\hat{\beta}_{ridge}(\lambda_{ridge})$